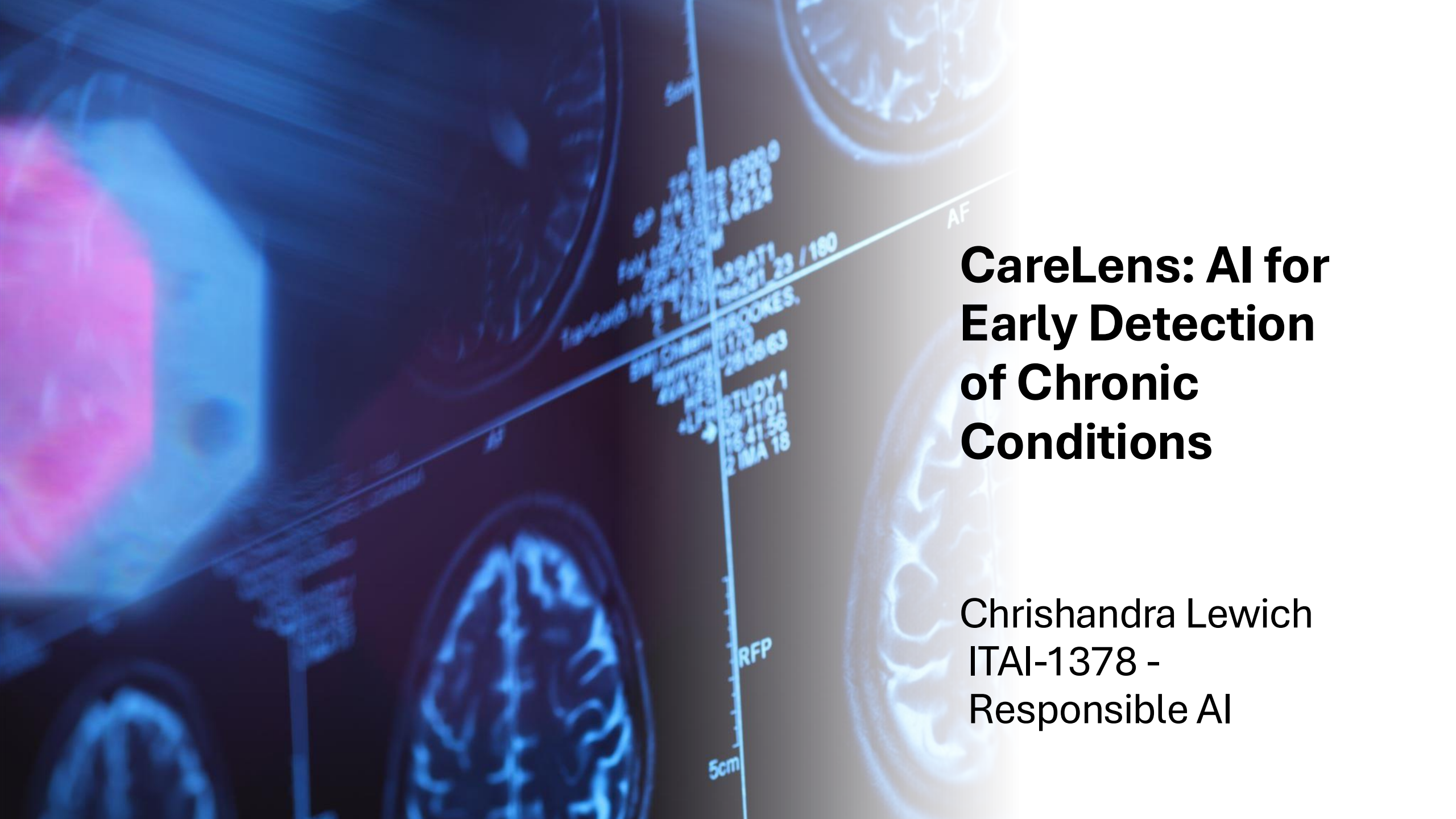


The background of the slide is a composite of medical scan images. On the left, there's a pinkish, textured area. The rest of the background is a dark blue field with several circular cross-sections of what appear to be brain or organ scans. Overlaid on this is a white, semi-transparent rectangular box containing technical text. The text includes patient information like 'PATIENT ID: 12345', 'DATE: 04/24', and 'STUDY: 1', as well as anatomical labels 'AF' and 'RFP'. A scale bar at the bottom left indicates '5cm'.

CareLens: AI for Early Detection of Chronic Conditions

Chrishandra Lewich
ITAI-1378 -
Responsible AI



CareLens is a hypothetical AI system that analyzes patient symptoms, medical records, and wearable device data to provide early alerts for chronic health conditions. This project examines CareLens from five perspectives:

- AI Ethicist
- Data Scientist
- Product Manager
- Consumer Advocate
- Government Regulator

Together, these viewpoints highlight the ethical, technical, and societal considerations required to responsibly develop AI in healthcare.

Product Overview

What CareLens Does

- Monitors health data from wearables and medical records
- Detects early signs of chronic conditions
- Sends alerts to patients and healthcare providers
- Supports preventive care and reduces hospitalizations

Users: Patients, doctors, clinics **Data Used:** Heart rate, sleep, blood pressure, glucose trends, medical history

AI Ethicist Perspective

Focus Areas:

- Fairness across demographics
- Transparency in predictions
- Privacy of sensitive health data
- Accountability for errors
- Respect for patient autonomy

Ethicist Position: CareLens must include fairness audits, explainable outputs, strict privacy controls, and clear responsibility guidelines.

Data Scientist Perspective

Focus Areas:

- High-quality, diverse training data
- Reliable model performance
- Secure data pipelines
- Continuous monitoring for false positives/negatives
- Appropriate feature selection

Data Scientist Position: CareLens can be effective with rigorous data practices, strong security, and ongoing evaluation.

Product Manager Perspective

Focus Areas:

- Market need for chronic disease management
- User adoption and ease of use
- Integration with existing wearables
- Risk management and user trust
- Regulatory compliance

Product Manager Position: CareLens has strong market potential but must balance innovation with safety and clarity.

Consumer Advocate Perspective

Focus Areas:

- User rights and informed consent
- Privacy and data protection
- Accessibility and affordability
- Prevention of discrimination
- Clear communication of risks

Consumer Advocate Position: CareLens must prioritize user protection, transparency, and equitable access.

Government Regulator Perspective

Focus Areas:

- Compliance with medical device laws
- HIPAA and data protection standards
- Clinical validation of AI predictions
- Liability and accountability rules
- Public interest and safety

Regulator Position: CareLens can be approved only with strong evidence of safety, fairness, and compliance.

Key Ethical Themes

- Bias and fairness
- Privacy and security
- Transparency and explainability
- Accountability and liability
- Societal impact and accessibility

These themes guide responsible AI development and deployment.

Conclusion

CareLens demonstrates how AI can support healthcare while raising important ethical, technical, and regulatory questions. Evaluating the system through five roles provides a balanced understanding of how responsible AI should be designed, tested, and governed.